STA 431/631 Introduction to Probability
Homework 5

Due: Saturday, October 24, 2026

Instructions: Complete the problems below. Show all your work for full credit.

Common Problems (Required for both STA 431 and STA 631)

1. Let X be a discrete random variable with the probability mass function (PMF):

$$f(x) = c(x^2 + x) \quad \text{for } x \in \{1, 2, 3, 4\}$$

   (a) Find the value of the constant c.

   (b) Calculate the expectation $E(X)$.

   (c) Calculate the variance $Var(X)$.

2. Let X be a continuous random variable with the probability density function (PDF):

$$f(x) = \begin{cases} \frac{3}{4}(1 - x^2) & \text{for } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Calculate the expectation $E(X)$ and the variance $Var(X)$.

   (b) Let $Y = X^2$. Find the expectation $E(Y)$ and the variance $Var(Y)$.

3. The continuous random variable X has the probability density function:

$$f(x) = \begin{cases} 3e^{-3x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Derive the moment generating function (MGF), $M_X(t)$. Be sure to state the condition on t for which the MGF exists.

   (b) Use the derivatives of the MGF to find the first moment $E(X)$ and the second moment $E(X^2)$.

4. The moment generating function (MGF) of a random variable X is given by $M_X(t) = e^{4t+8t^2}$. Let Z be a new random variable defined by the linear transformation $Z = \frac{X-4}{4}$.

   (a) Use the properties of MGFs to find the MGF of Z, $M_Z(t)$.

   (b) Use $M_Z(t)$ to find the expectation $E(Z)$ and the variance $Var(Z)$.

5. An insurance policy pays a total medical benefit consisting of two parts for each claim. Let X represent the part paid to the surgeon, and Y represent the part paid to the hospital. The variance of X is 300, the variance of Y is 500, and the covariance between X and Y is 150.

   Due to rising costs, the company updates the policy. The new payout to the surgeon will be $W = 2X + 100$, and the new payout to the hospital will be $V = 1.1Y$. Calculate the variance of the new total benefit, $Var(W + V)$.

6. Let X, Y, and Z be random variables. We are given the following variances and covariances: $Var(X) = 4, Var(Y) = 9, Var(Z) = 2, Cov(X, Y) = 1, Cov(X, Z) = -1$, and $Cov(Y, Z) = 0$. Calculate the variance of the linear combination $W = 3X - 2Y + Z$.

7. Let the continuous random variables X and Y have the joint PDF:

$$f(x, y) = \begin{cases} x + y & \text{for } 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginal PDF of Y, $f_Y(y)$.

   (b) Derive the conditional PDF of X given $Y = y$, $f_{X|Y}(x \mid y)$.

   (c) Find the conditional mean $E(X \mid Y = 1/2)$ and the conditional variance $Var(X \mid Y = 1/2)$.

STA 431 Only Problems

8. Let the joint PMF of X and Y be defined by:

$$f(x, y) = \begin{cases} \frac{x+y}{36} & \text{for } x \in \{1, 2, 3\} \text{ and } y \in \{1, 2, 3\} \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginal expectations $E(X)$ and $E(Y)$.

   (b) Find the product moment $E(XY)$.

   (c) Calculate the covariance $Cov(X, Y)$.

   (d) Find the conditional PMF $f_{Y|X}(y \mid 1)$ and calculate the conditional expectation $E(Y \mid X = 1)$.

STA 631 Only Problems

8. Let X be a continuous random variable with PDF $f(x) = 1 - |x|$ for $-1 < x < 1$, and 0 otherwise. Let $Y = X^2$.

   (a) Prove that the covariance $Cov(X, Y) = 0$.

   (b) Are X and Y independent? Provide a formal mathematical justification for your answer.
   (Hint: Recall that X and Y are independent if and only if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all sets A and B. To show they are not independent, construct an explicit pair of intervals A and B of positive probability for which this equality fails.)